# wheat sourdough

*White wheat flour + 25% whole wheat flour. Stiff starter, lievito madre at 43% hydration — my standard, super-active one, the starter I baked bread and pastry on at my bakery. Using a different starter? Don't worry, the recipe can be adapted! Further down the recipe there's a calculator that converts it to a liquid starter and to any amount of flour and number of loaves. Dough hydration in this recipe is 75%. Sounds scary, it's simple.*

| 75% | 43% |
|---|---|
| HYDRATION | STARTER |

**1 night**
COLD PROOF



white + whole wheat

- [ ] scale — MUST
- [ ] bowl + spoon — MUST
- [ ] basket *or bowl + towel* — SWAP
- [ ] cast iron *or roasting bag* — SWAP
- [ ] blade *or sharp knife* — SWAP
- [ ] parchment + mitts — MUST
- [ ] rack *or the oven rack* — SWAP
- [ ] thermometer — NICE

# what you need

*Tick what you already have before you start. Almost everything has an easy swap — except the scale.*



# what goes in

| ingredient | small loaf 600 G | big loaf 1150 G |
|---|---|---|
| starter · 43% | 71 g | 137 g |
| whole wheat flour | 85 g | 163 g |
| white bread flour | 204 g | 391 g |
| water | 233 g | 446 g |
| salt | 7 g | 13 g |

*Starter 43% = 21% of the flour. Inside it: 50 g flour + 21 g water (600 g loaf), 96 g flour + 41 g water (1150 g loaf).*

**11%+**

PROTEIN · WHITE FLOUR

**10%+**

PROTEIN · WHOLE WHEAT

*Mine is Caputo Nuvola, 12.5%. Tools that matter: a scale, cast iron or any covered pot, and a blade for scoring.*



## liquid starter 100%?

a — Swap so the flour and water in the dough stay the same.

b — 600 g: starter 100 g (not 71), water 204 g (not 233). Everything else the same.

c — 1150 g: starter 192 g (not 137), water 391 g (not 446). Everything else the same.

d — Any other starter: it must carry 50 g / 96 g of flour; cut the dough water by whatever it brings above 21 g / 41 g. Or let the calculator do it.

# 01 mix & walk away

a — Weigh the water. Tear the starter into pieces — it gets to work faster.

b — White flour in, then whole wheat.

c — Mix with a spatula, just to wet the flour. No gluten development yet.

d — Cover and leave for autolyse.

**30–50**
MIN AUTOLYSE

## just wet the flour →

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s1

# 02 salt & the important part

a Salt in. Now knead properly.

b Taste from two spots. One bland, one salty? Keep kneading.

c At the end, measure the dough temperature. Mine was 27 °C.

d In this recipe my working range is 23–25 °C. Above 26 °C fermentation in my process runs ahead of the gluten, so I cool the dough — 30 min in the fridge.

| 2% | 26° | 23–25° |
| --- | --- | --- |
| SALT OF FLOUR | ABOVE IT — COOL DOWN | MY WORKING RANGE |

dough too warm? use cold water next time →

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s2

# 03 back in the optimal zone

a Out of the fridge: the dough is 25 °C. A great temperature for fermentation in the apartment.

b Next fold: wet hands, stretch one side up and fold it over, turn the bowl, repeat on all four sides.

c Cover and leave for 30 min until the next fold.

**25°**
AFTER THE FRIDGE

**23–25°**
MY WORKING RANGE

*a calm dough, right on track →*

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s3

# 04 folds & waiting

a A fold every 30 minutes, four in total.

b By the third one you see bubbles. A good sign.

c On the fourth, check the gluten: stretch a piece thin. A window like this is what we want.

d Now wait until the dough grows by 50–60%. Mine took about an hour and a half. And please forgive the sunset light in the last shots! I beeeg you

**4**
FOLDS · EVERY 30 MIN

**+50–60%**
VOLUME, NOT TIME

*watch the dough, not the clock →*

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s4

# 05 shape & say good night

a Preshape: round it, tightening the top. Rest 20–30 min.

b Final shape: I do folds and swaddling. Any method you like works.

c Another 20–30 min proof, then into the fridge overnight, covered.

d My fridge is about 7 °C. At that temperature fermentation keeps going, so the overnight timing is for my fridge and this process: in a colder fridge the dough can stay longer, in a warmer one bake it earlier.

**20–30**
MIN REST · ×2

**~7°**
MY FRIDGE, OVERNIGHT

*tension on top, not force →*

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s5

# 06 good morning. let's bake

a Morning. The dough looks like this — time to bake.

b Turn it out onto a mat, score on the side.

c The dutch oven has been heating for an hour at 250 °C.

d Lid on — 20 min. Lid off, down to 220 °C — another 20-25.

| 250° | 220° |
| --- | --- |
| 20 MIN, LID ON | 20-25 MIN, LID OFF |

*the score opened a bit weakly? no worries! with a big loaf that's completely normal! →*

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s6

TIMING AT A GLANCE — WATCH THE DOUGH, NOT THE CLOCK

a Autolyse 30-50 min → salt, knead → temperature check: above 26 °C, cool to 23-25 °C

b 4 folds, every 30 min → bulk until +50-60%

c Preshape, rest 20-30 min → shape, proof 20-30 min → fridge overnight (~7 °C)

d Bake 250 °C lid on 20 min + 220 °C lid off 20-25 min

*Timing is approximate. A liquid starter may run a little faster or slower than lievito madre.*

## TROUBLESHOOTING

IF SOMETHING WENT WRONG

| problem | possible cause |
|---|---|
| The loaf spread flat | *Not enough structure developed, or overproofed* |
| Dense crumb | *Underfermented* |
| Too sour | *Starter ripeness and how the fermentation went* |
| Crust too thick | *Baking mode and moisture loss* |
| The loaf barely opened | *Shaping, proofing or scoring* |

## NUTRITION

APPROXIMATE, PER 100 G OF BREAD

| 228 KCAL | 8.4 g PROTEIN | 1.2 g FAT | 47 g CARBS |
|---|---|---|---|

a A slice of about 40 g is around 90 kcal. The whole loaf is about 2230 kcal · protein 82 g · fat 11 g · carbs 458 g.

b Only the flour counts: 487 g white (391 g in the dough + 96 g inside the starter) and 163 g whole wheat. Water and salt have no calories.

c Estimate for a loaf that lost about 15% of its weight in the oven (1150 g of dough → about 980 g of bread). Weigh yours and divide the whole-loaf numbers by its weight for an exact figure.

d Carbs include about 2.5 g of fibre. Salt: about 1.3 g per 100 g.



the ear



listen to that crust



one hour later

## and... here it is

*Here is our bread. The score opened a bit weakly — that's the volume. In homemade bread that's not the point. The point is taste, craft and love.*



taste, craft & love